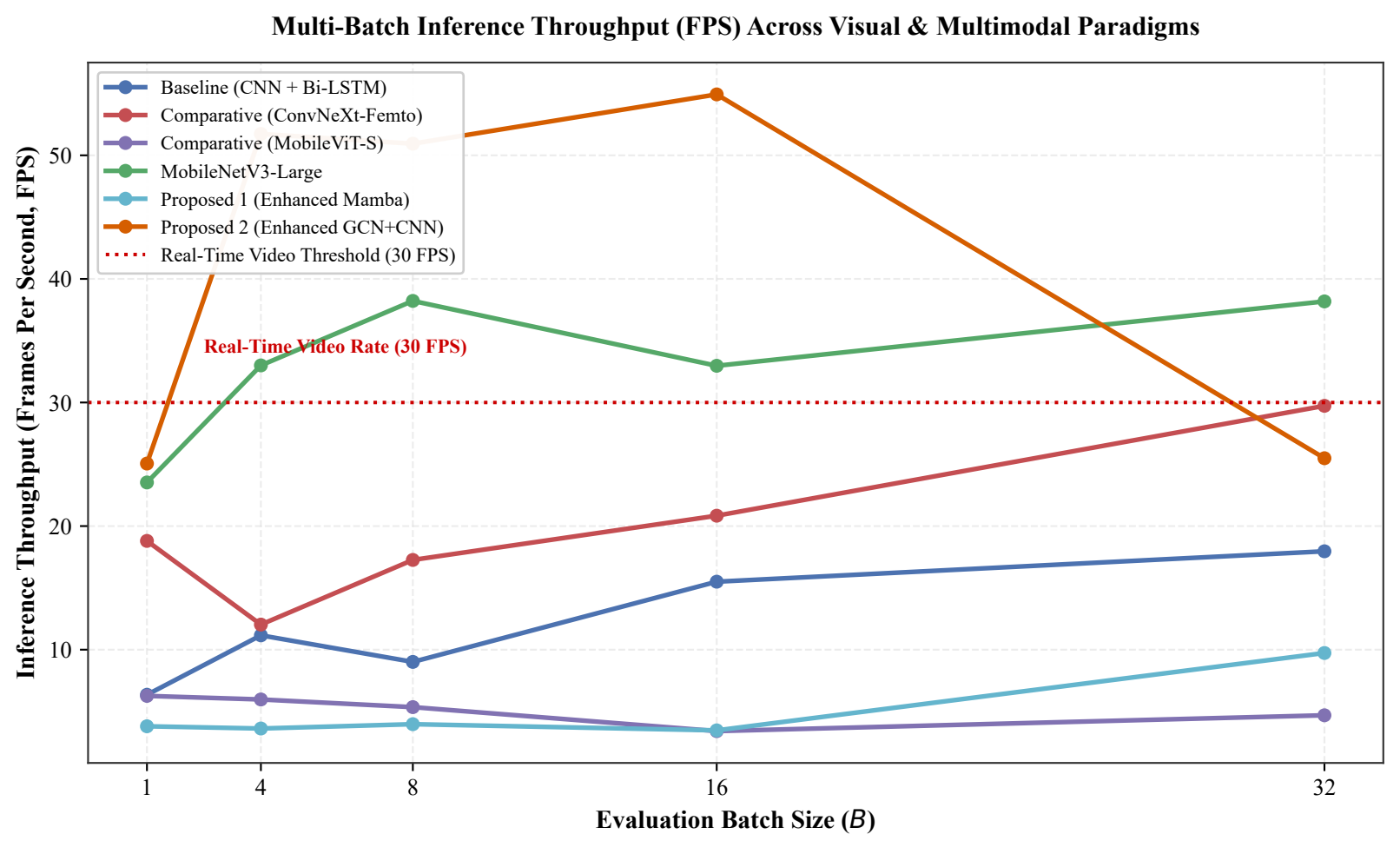

Multi-Batch Inference Throughput (FPS) Across Visual & Multimodal Paradigms
Baseline (CNN + Bi-LSTM)
Comparative (ConvNeXt-Femto)
Comparative (MobileViT-S)
MobileNetV3-Large
Proposed 1 (Enhanced Mamba)
Proposed 2 (Enhanced GCN+CNN)
Real-Time Video Threshold (30 FPS)
Real-Time Video Rate (30 FPS)
Inference Throughput (Frames Per Second, FPS)
50
40
30
20
10
1
4
8
16
32
Evaluation Batch Size (B)